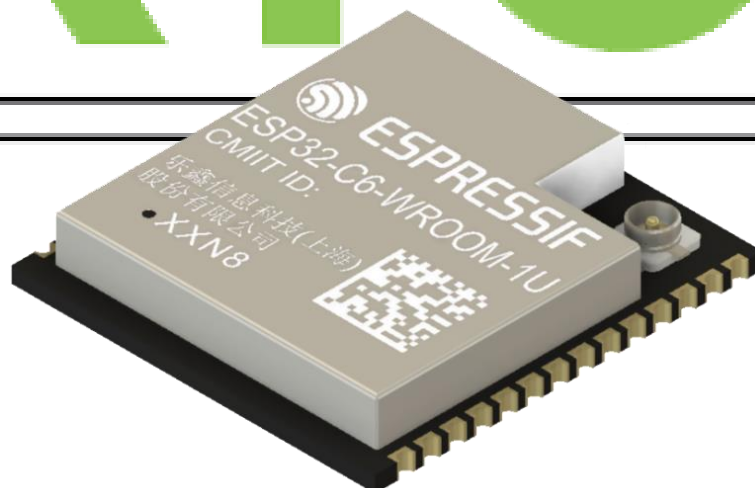


30-Day FreeRTOS Course for ESP32 Using ESP-IDF

(Day 18)

“Software Timers”

freeRTOS



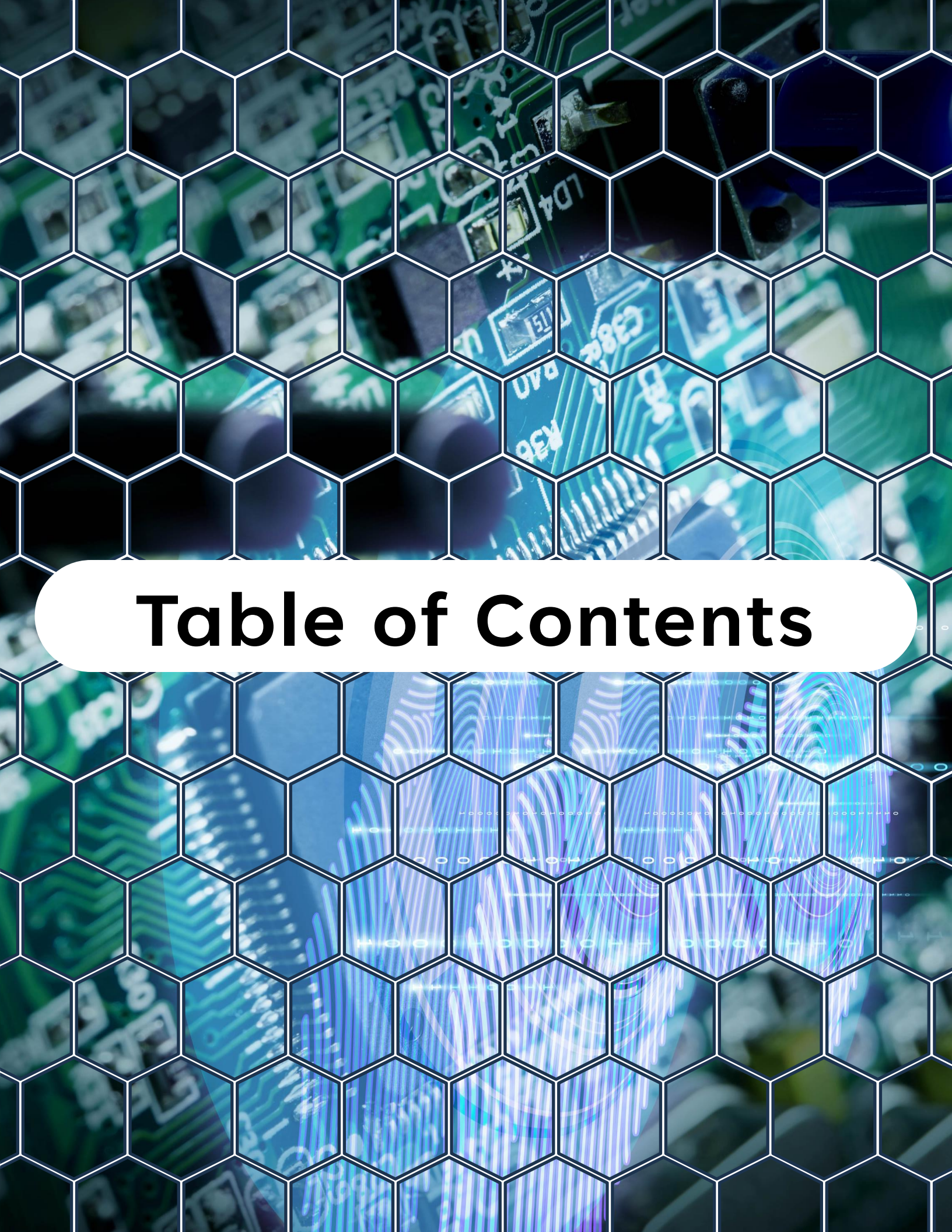
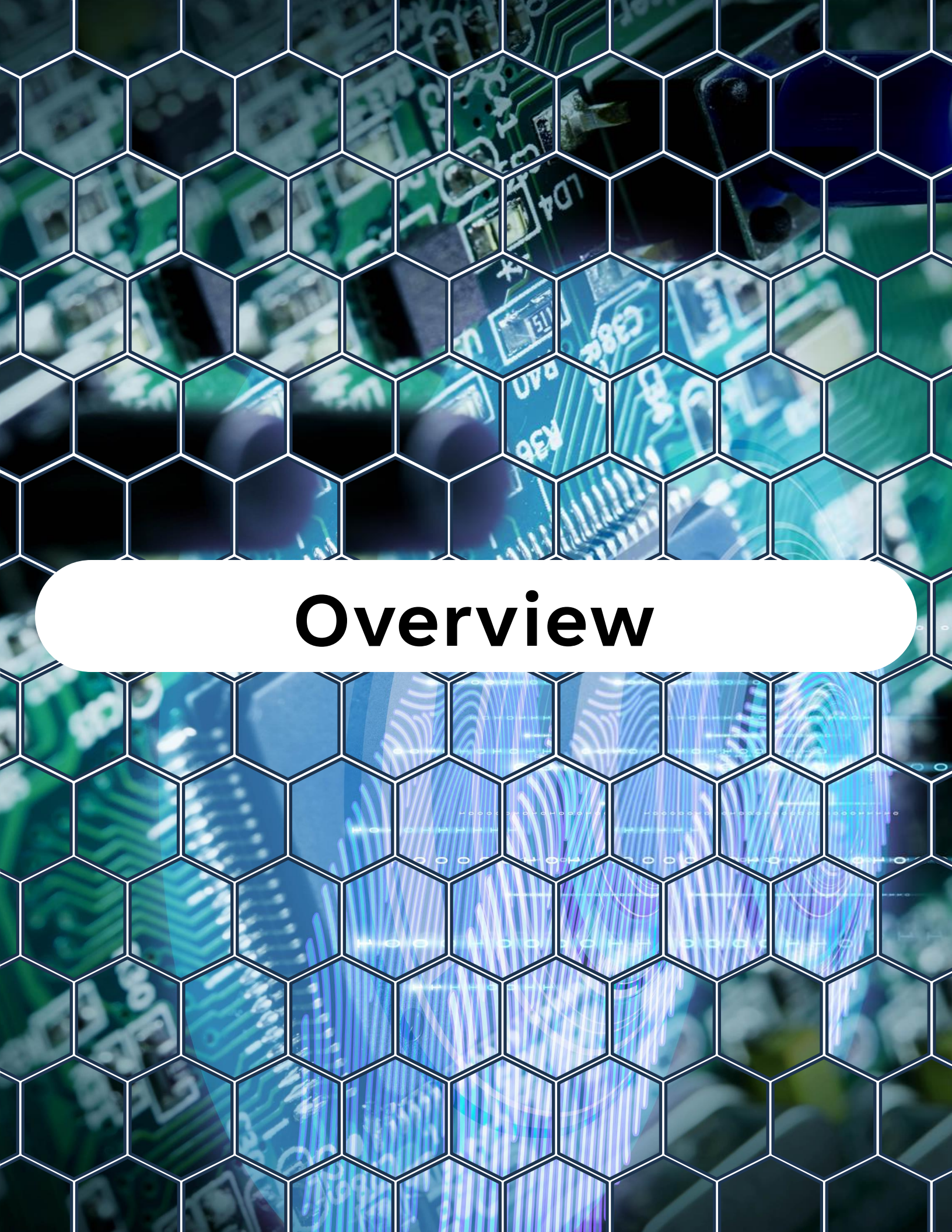


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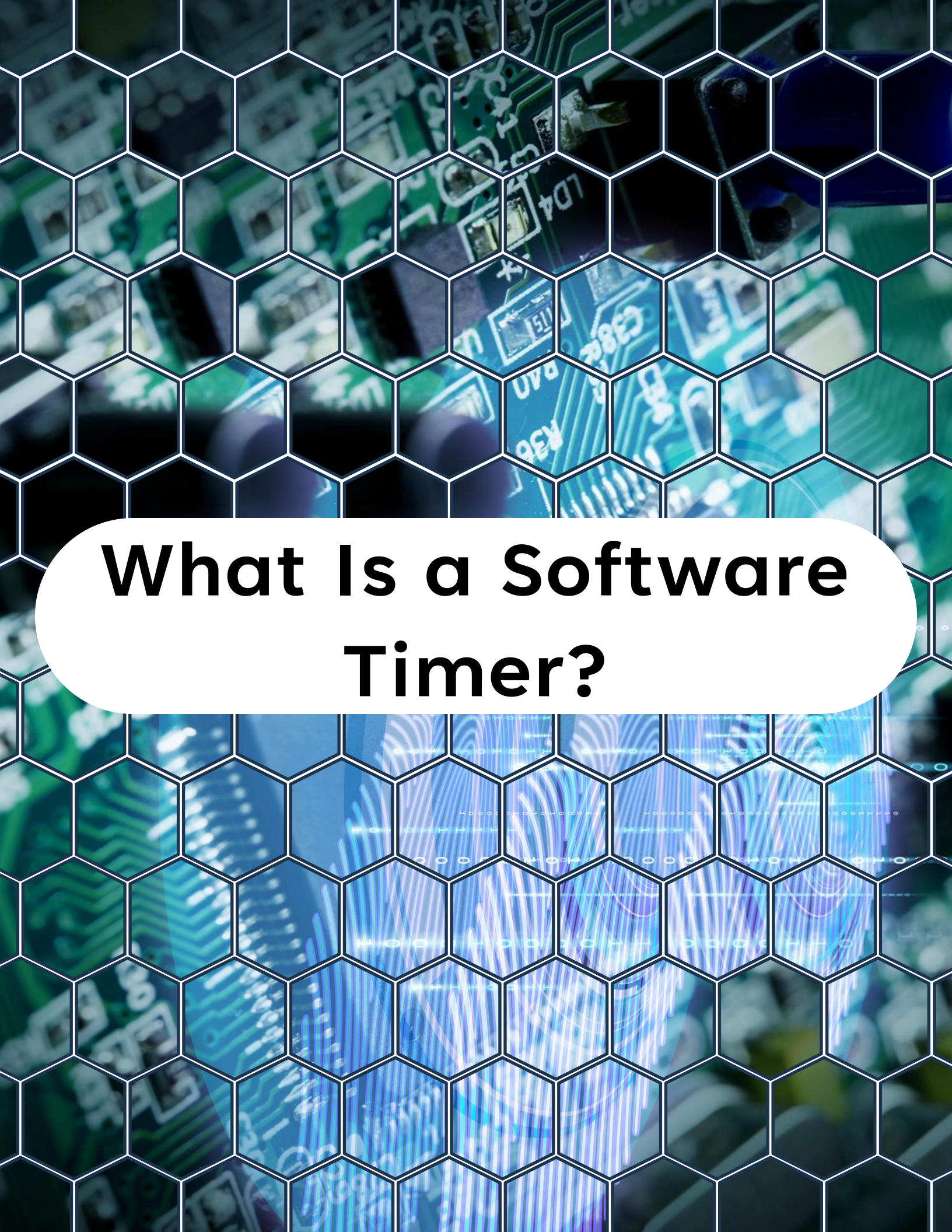
Overview

Overview

On **Day 18**, you'll learn:

- What **software timers** are in FreeRTOS
- The difference between **one-shot** and **periodic timers**
- How to create, start, stop, and reset timers
- How software timers differ from hardware timers
- A practical ESP32 example using timers for periodic tasks
- Best practices for software timer usage

By the end of this lesson, you'll be able to use **FreeRTOS software timers** for deferred execution and periodic actions without creating dedicated tasks.



What Is a Software Timer?

What Is a Software Timer?

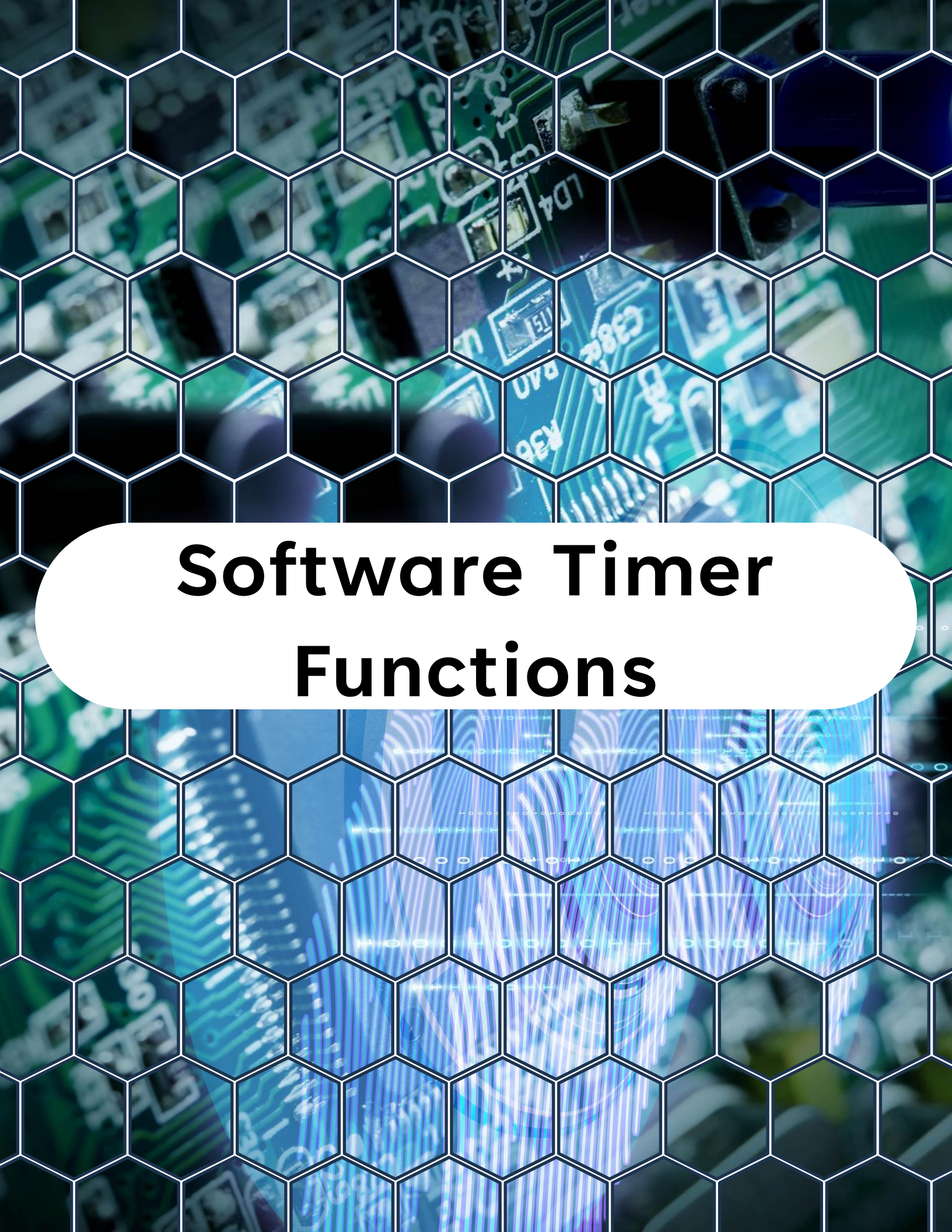
A **software timer** is a FreeRTOS mechanism that executes a **callback function** after a specified period.

Two types:

- **One-shot timer** → runs once, then stops.
- **Periodic timer** → automatically restarts after expiration.

Software timers run in the **Timer Service Task** (created internally by FreeRTOS). This means:

- Callbacks **don't run in ISR context** (safe to use most FreeRTOS APIs).
- Callbacks must be **short and efficient** to avoid delaying other timers.



Software Timer Functions

Software Timer Functions

Create a Timer



```
1 TimerHandle_t xTimerCreate(  
2     const char *pcTimerName,  
3     TickType_t xTimerPeriodInTicks,  
4     UBaseType_t uxAutoReload,  
5     void *pvTimerID,  
6     TimerCallbackFunction_t pxCallbackFunction);
```

- `xTimerPeriodInTicks` → timer period in ticks.
- `uxAutoReload` → `pdTRUE` for periodic, `pdFALSE` for one-shot.
- `pvTimerID` → optional ID passed to callback.
- `pxCallbackFunction` → function executed when timer expires.

Software Timer Functions

Start/Stop/Reset



```
1 xTimerStart(TimerHandle_t xTimer,  
2             TickType_t xTicksToWait);  
3 xTimerStop(TimerHandle_t xTimer,  
4            TickType_t xTicksToWait);  
5 xTimerReset(TimerHandle_t xTimer,  
6            TickType_t xTicksToWait);
```

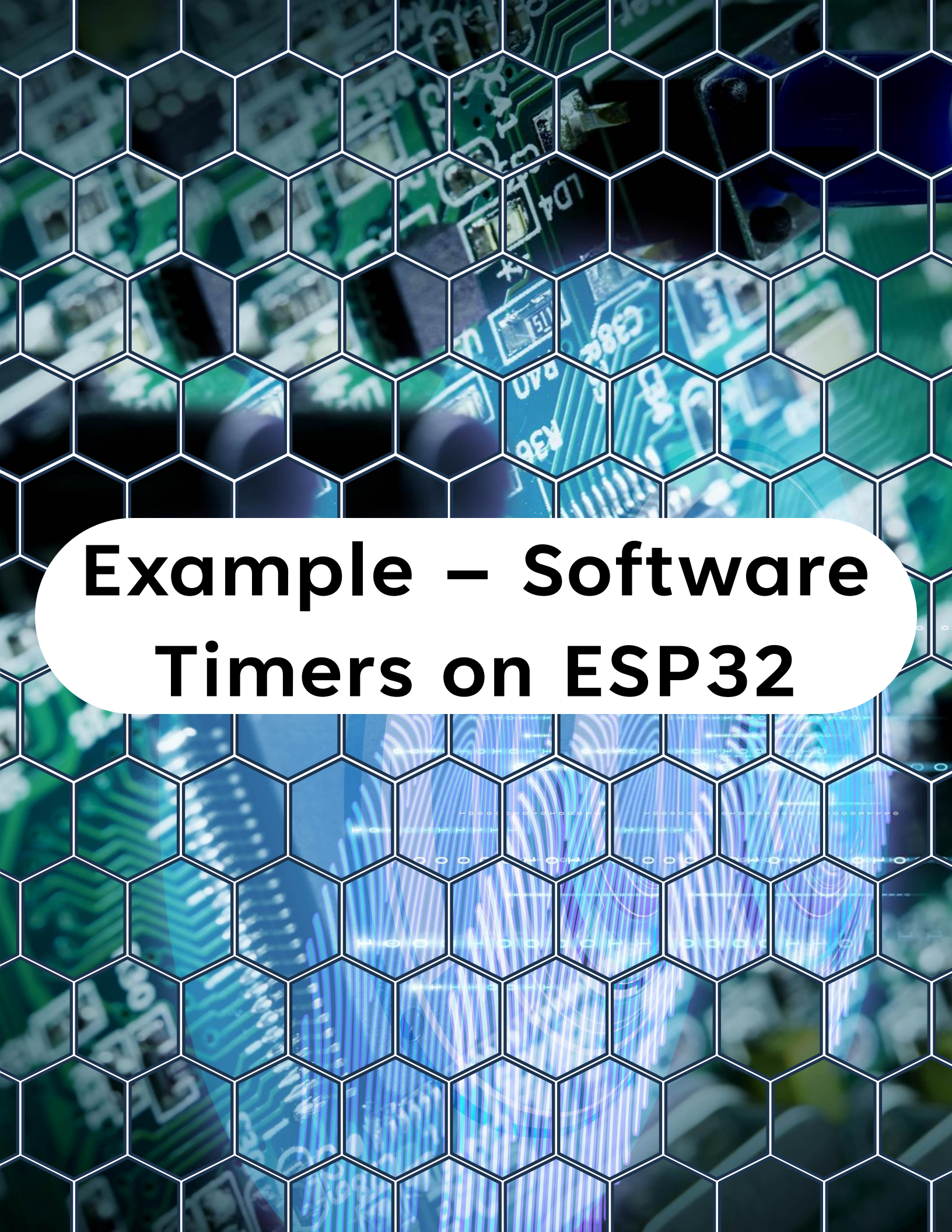
ISR-safe versions exist (e.g., `xTimerStartFromISR()`).



Use Cases

Use Cases

- **LED blink without a dedicated task**
- **Watchdog-style timeouts**
- **Retry logic (e.g., reconnect Wi-Fi every 5s)**
- **Periodic monitoring (e.g., sensor polling)**



Example – Software Timers on ESP32

Example – Software Timers on ESP32

Code Example



```
1  #include <stdio.h>
2  #include "freertos/FreeRTOS.h"
3  #include "freertos/timers.h"
4  #include "esp_log.h"
5
6  #define TAG "DAY18"
7
8  // Timer handles
9  TimerHandle_t one_shot_timer;
10 TimerHandle_t periodic_timer;
11
12 // Callback for one-shot timer
13 void one_shot_callback(TimerHandle_t xTimer) {
14     ESP_LOGI(TAG, "One-shot timer expired!");
15 }
16
17 // Callback for periodic timer
18 void periodic_callback(TimerHandle_t xTimer) {
19     static int count = 0;
20     count++;
21     ESP_LOGI(TAG, "Periodic timer tick %d", count);
22
23     if (count == 5) {
24         ESP_LOGI(TAG, "Stopping periodic timer after 5 ticks");
25         xTimerStop(xTimer, 0);
26     }
27 }
28
29 // Main application entry point
30 void app_main() {
31     // Create timers with 3s interval (one-shot and periodic)
32     one_shot_timer = xTimerCreate("OneShot",
33                                   pdMS_TO_TICKS(3000), // 3s delay
34                                   pdFALSE,
35                                   // One-shot
```

Example – Software Timers on ESP32

```
29 // Main application entry point
30 void app_main() {
31     // Create timers with 3s interval (one-shot and periodic)
32     one_shot_timer = xTimerCreate("OneShot",
33                                   pdMS_TO_TICKS(3000), // 3s delay
34                                   pdFALSE,             // One-shot
35                                   NULL,
36                                   one_shot_callback);
37
38     // Create periodic timer with 1s interval
39     periodic_timer = xTimerCreate("Periodic",
40                                   pdMS_TO_TICKS(1000), // 1s interval
41                                   pdTRUE,               // Auto-reload
42                                   NULL,
43                                   periodic_callback);
44
45     // Start timers
46     xTimerStart(one_shot_timer, 0);
47     xTimerStart(periodic_timer, 0);
48 }
```


Example – Software Timers on ESP32

Expected Behavior

After 3 seconds, the one-shot timer expires and logs:

I (3000) DAY18: One-shot timer expired!

Every 1 second, the periodic timer logs:

I (1000) DAY18: Periodic timer tick 1

I (2000) DAY18: Periodic timer tick 2

...

I (5000) DAY18: Stopping periodic timer after 5 ticks



Best Practices

Best Practices

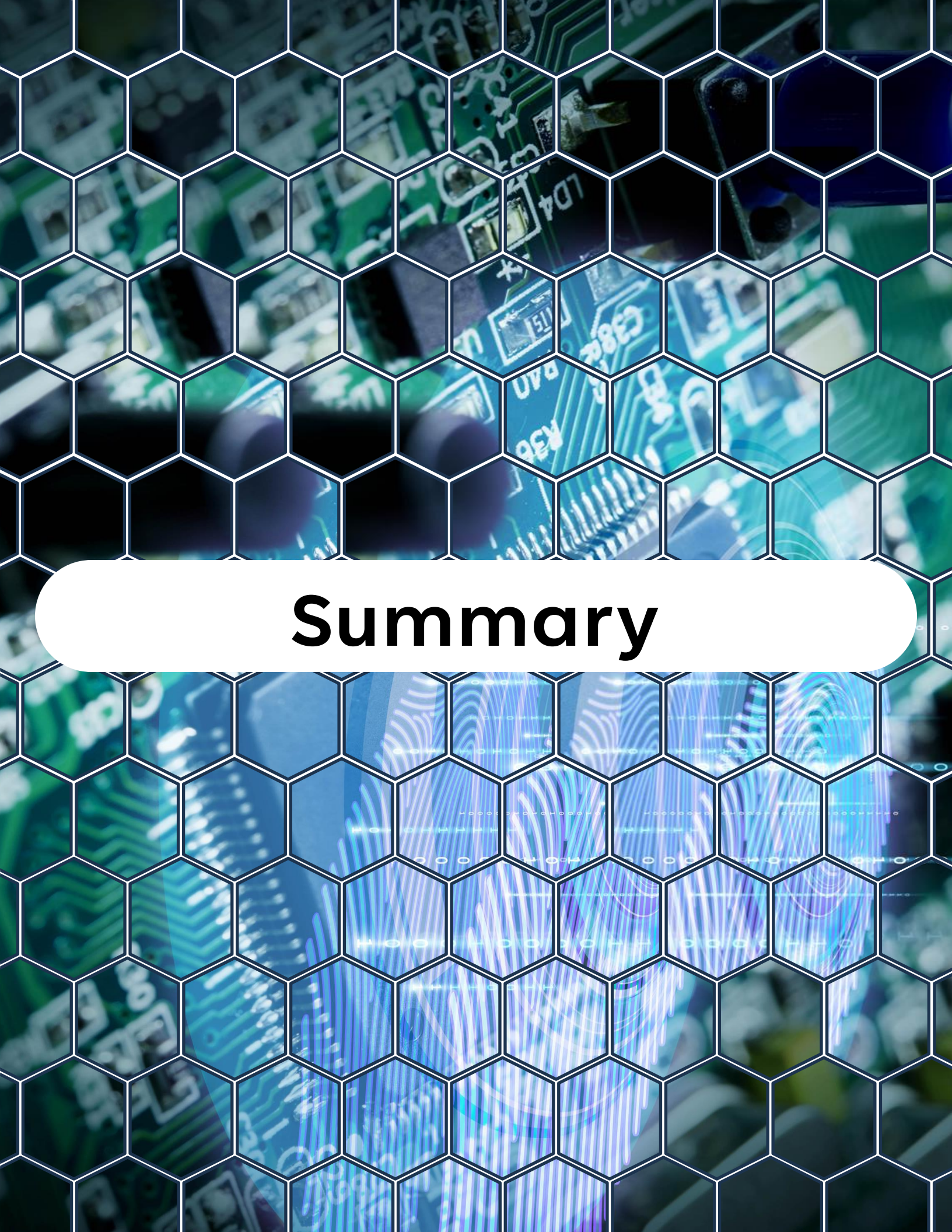
✓ Use software timers to replace dedicated “delay tasks” → more efficient.

✓ Keep callbacks **short and non-blocking** (never call `vTaskDelay()` inside).

✓ For frequent, high-resolution timing, consider **hardware timers**.

✓ Use `xTimerReset()` to restart timers for **timeout management**.

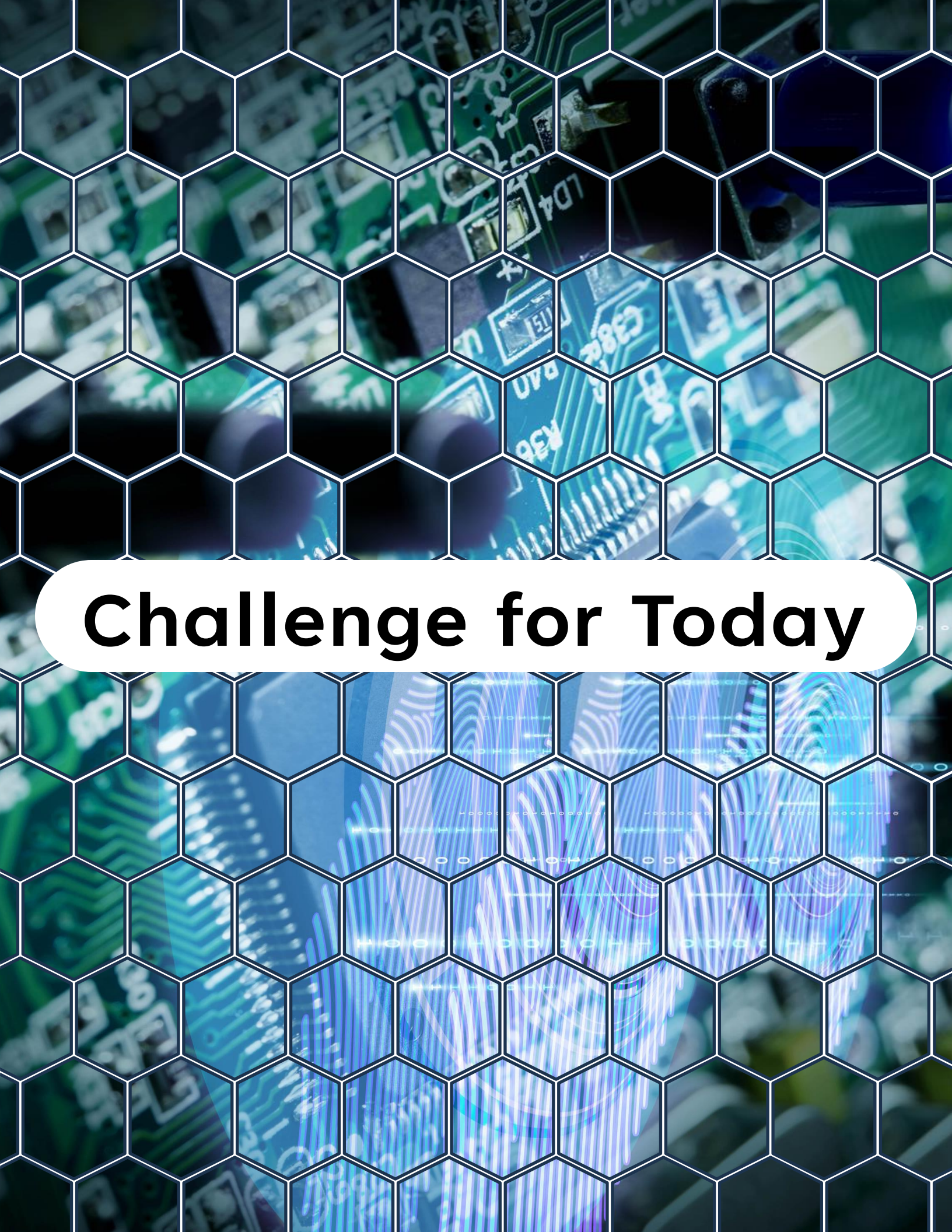
✓ Don't rely on timers for **hard real-time** deadlines (jitter can occur).



Summary

Summary

- Software timers allow **one-shot or periodic callbacks** without extra tasks.
- They're efficient for **timeouts, retries, and periodic work**.
- Managed by the **FreeRTOS Timer Service Task**.
- Keep callbacks short, safe, and non-blocking.



Challenge for Today

Challenge for Today



Create two timers:

- A **5s one-shot timer** that toggles an LED once.
- A **500ms periodic timer** that toggles the LED repeatedly.
- Stop the periodic timer after 10 toggles using a counter in the callback.

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